



Large eddy simulation of piloted turbulent partially cracked ammonia flames with the direct moment closure model

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ABSTRACT

Ammonia is considered as a promising carbon-free fuel. The direct moment closure (DMC) model, which we previously developed for turbulence-chemistry interactions modeling (Luo et al., Proc. Combust. Inst., 2021), directly closes the filtered chemical reaction source term. It does not need to introduce the flame mode assumptions and has been verified on flames of hydrocarbon fuels. However, its effectiveness for ammonia combustion modeling needs further verification. To this end, the objective of the current study is to further evaluate the performance of the DMC model in predicting turbulent ammonia combustion. Large eddy simulations (LESs) of the KAUST piloted jet flames of partially cracked ammonia are performed. In particular, the target flames D and F with different Reynolds numbers are studied. Quantitative comparisons are made between model predictions and experimental measurements in the physical space and the mixture fraction space. Local extinction within the combustion zone is emphasized in the discussion. The results show that LES-DMC can accurately reproduce the thermo-chemical states in the stable combustion region through comparisons in the physical and mixture fraction spaces, while deviations exist in local areas with significant extinction phenomena. Overall, the DMC model is a promising model for turbulent ammonia combustion, and further improvements are needed to describe unsteady phenomena more accurately.

1. Introduction

Ammonia (NH_3), as a carbon-free fuel, is gaining growing interest for its capability to serve as a substitute for conventional fossil fuels. However, the utilization of ammonia for combustion encounters challenges due to its low heat value and low laminar burning velocity [1,2]. Partial cracking of ammonia is an effective strategy that can produce hydrogen to enhance combustion stability without the need for additional fuel [3].

Large eddy simulation (LES) can provide detailed physical and chemical information in turbulent ammonia/hydrogen combustion. To capture the complex turbulence-chemistry interactions within the framework of LES, turbulent combustion modeling is required. Detailed experimental measurements of turbulent flames can guide and validate the turbulent combustion modeling. A few experimental studies have been conducted to perform quantitative measurements on turbulent ammonia/hydrogen flames at the laboratory scale. Tang et al. [4] measured scalar structure in turbulent non-premixed ammonia/hydrogen/nitrogen jet flames, offering the first dataset for validation of numerical models. Recently, Tang et al. [5] have further carried out an experimental study on piloted turbulent partially premixed ammonia/hydrogen/nitrogen-air jet flames. They observed significant

local extinction phenomena in the flame with higher Reynolds number. Additionally, they quantitatively measured temperature, mixture fraction, and OH concentration using 1D Raman/Rayleigh/LIF techniques, providing benchmark data for examining numerical models.

A variety of turbulent combustion models have been developed for coupling with large eddy simulation, such as the flamelet-like models [6–8], PDF-like models [9,10], G-equation [11], conditional moment closure [12], thickened flame model [13] and so on. However, most of these models rely on assumptions about the premixed or non-premixed combustion regimes. Studies have shown that even in the flames traditionally considered to be premixed or non-premixed, both premixed and non-premixed conditions may coexist [14,15], which poses challenges for combustion modeling. In our previous studies, we proposed a direct moment closure (DMC) model [16,17], which was extended from the second-order moment [18] and the dynamic second-order moment closure [19] models. In the DMC model, the reaction rate source term is directly closed by the Arrhenius law without introducing any assumptions about the combustion mode. Therefore, the DMC model is expected to be applicable to premixed, non-premixed, and partially premixed conditions. In our previous studies, the DMC

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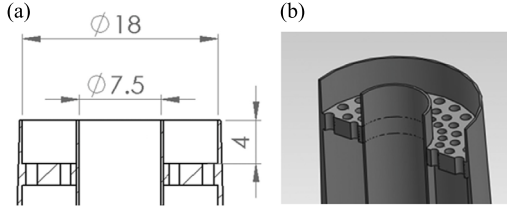


Fig. 1. The studied burner [5,20], (a) nozzle dimensions in mm, (b) geometric illustration.

model has been validated for flame configurations of conventional hydrocarbon fuels, such as methane [16] and heptane [19]. However, its effectiveness in the combustion of low-carbon fuels such as ammonia still needs to be verified.

To this end, the present work aims to further assess the performance of the DMC model in predicting turbulent partially cracked ammonia combustion. LESs of the KAUST piloted ammonia/hydrogen/nitrogen-air jet flames (Flames D and F) [5], which are identified as target flames in the TNF workshop, are conducted using the DMC model, and the simulation results are compared with the experimental data in both the physical and mixture fraction spaces.

2. Experimental setup

The experimental burner is a modified pilot-assisted jet burner based on the Sydney in-homogeneous jet burner [5,20]. The modification is removing the central recessed fuel pipe. As shown in Fig. 1, the modified burner has a central fuel jet with an inner diameter $D = 7.5$ mm, surrounded by an annular pilot region with an inner diameter of 18 mm. A surrounding coflow of air with a diameter of 250 mm and a velocity of 0.8 m/s is stably delivered outside the pilot region.

Two flames designated as Flame D and Flame F are studied. Both flames utilize a fuel mixture designed to simulate 43% cracked NH_3 . This mixture is further premixed with air to reach an equivalence ratio $\phi = 3$. Additionally, both flames employ a pilot composition consisting of 43% cracked NH_3 and air with $\phi = 0.9$. The key difference between the flames is the Reynolds number Re : Flame D operates at $Re = 24,000$ (59% of the extinction Re) while Flame F has a higher Re of 36,000 (89% of extinction Re). Higher Reynolds numbers correspond to elevated levels of turbulence, increased shear stress, and consequently higher scalar dissipation rates within the reaction zone. Therefore, with a higher Reynolds number, there is a higher probability of local extinction occurring in Flame F compared to Flame D.

The experimental study [5] adopted a 1D combined system of Raman/Rayleigh scattering and NH_2/OH laser-induced fluorescence (LIF) to measure major species, mixture fraction, temperature, and OH mole fraction, along with semi-quantitative NH_2 -LIF detection. Data is collected at various axial locations along the flame.

3. Direct moment closure model

For a detailed description of the direct moment closure model, refer to our previous studies [16,17]. Here, we briefly outline the main modeling strategies. Generally, the rate of a chemical reaction is determined using the Arrhenius law. However, in the framework of large eddy simulation, the filtered reaction rate needs to be modeled. In the DMC model, the filtered reaction rate is represented as the product of a constant term $B = A \prod_{i=1}^N W_i^{-\nu_i}$, a temperature term $K = T^\beta \exp(-E/RT)$, and a concentration term $R_i = \rho Y_i$ as below:

$$\bar{\dot{\omega}} = AT^\beta \exp(-E/RT) \prod_{i=1}^N \left(\frac{\rho Y_i}{W_i} \right)^{\nu_i} = BK \prod_{i=1}^N R_i^{\nu_i}, \quad (1)$$

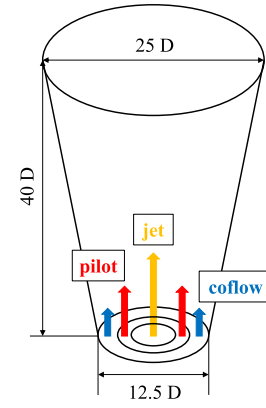


Fig. 2. Schematic of the computational domain.

where A , β and E denote the pre-exponential factor, the temperature exponent and the activation energy of the reaction, respectively. R is the universal gas constant, N is the number of reactants and ν_i is the concentration exponent of species i . For a typical double-reactant elementary reaction, the filtered reaction rate can be rewritten as the sum of a first-order moment, several second-order moments, and a third-order moment. In some previous studies [19,21], the third-order moment was considered to have a minor impact and was neglected. However, this assumption is not always valid [16]. To avoid introducing this assumption, a new formula was proposed in our previous study [17]. Through mathematical transformation, the filtered reaction rate can be split into an average term and two fluctuation terms. Only the temperature term $\bar{K}$ as well as the two fluctuation terms $\bar{R}_1 \bar{R}_2 - \bar{R}_1 \bar{R}_2$ and $\bar{K} \bar{R}_1 \bar{R}_2 - \bar{K} \bar{R}_1 \bar{R}_2$ require modeling. According to previous studies, a locally linearized model is used to represent the temperature term $\bar{K}$ [19], while an algebraic closure model [22] is employed to calculate the fluctuation terms:

$$\overline{\phi' \phi'} = \overline{\phi \phi} - \overline{\phi} \overline{\phi} = C_{\phi\phi} \Delta^2 \frac{\partial \bar{\phi}}{\partial x_j} \frac{\partial \bar{\phi}}{\partial x_j}, \quad (2)$$

where the coefficient $C_{\phi\phi}$ is determined using the classic dynamic model (CDM) [22]. These models have been *a priori* validated using DNS data in previous studies [16,19]. Based on the above modeling approach, the DMC model has a simple formulation, is easy to implement in LES solvers and does not introduce any assumptions about the combustion mode.

4. Numerical setup

The DMC model has been implemented in the open-source C++ platform OpenFOAM [23] and coupled with an existing low-Mach number compressible LES solver to perform LES-DMC simulations. The sub-grid stress is modeled using a dynamic Smagorinsky model [24]. As shown in Fig. 2, the computational domain is a cylindrical domain with variable diameter, ranging from $12.5 D$ to $25 D$, and a length of $40 D$. Consistent with the experiment, the inlet boundary consists of three streams: jet, pilot, and coflow. The inlet velocity of the pilot is set to 6.55 m/s, which is calculated based on the velocity of the fresh pilot flow and the mass conservation. The inlet velocity of the coflow is set to 0.8 m/s. The bulk jet velocities of Flame D and Flame F are 61.07 m/s and 91.6 m/s, respectively. The specific velocity distributions of the jet are generated using the divergence free synthetic eddy method (DFSEM) [25] for rapid turbulent development. The inlet boundary conditions for the species and temperature are listed in Table 1. The boundaries for the jet and coflow are consistent with the experiment, and the boundary for the pilot is obtained from the adiabatic calculation of a mixture of 43% cracked ammonia and air at an equivalence ratio of 0.9. The outlet employs an “inletOutlet”

Table 1

Temperature and mass fractions of the jet, pilot, and coflow inlets.

Parameters	Jet	Pilot	Coflow
T [K]	293.5	2047	295.6
Y_{O_2} [-]	0.1574	0.0203	0.233
Y_{N_2} [-]	0.6394	0.7746	0.767
Y_{NH_3} [-]	0.1770	0	0
Y_{H_2} [-]	0.0262	0	0
Y_{H_2O} [-]	0	0.2033	0
Y_{OH} [-]	0	0.0018	0

condition, which switches between a zero-gradient and a fixed-value condition based on the flow direction. A skeletal mechanism with 31 species and 203 reactions from the work of Stagni et al. [26] is used to model the chemical kinetics. For a preliminary validation, the simplified assumption of unity Lewis number is used for molecular diffusion. Given that the target flames contain hydrogen, the potential limitations of this assumption will be discussed later. To compare with the experiment, the mixture fraction defined as Eq. (3) is calculated at each time step, where Y_i and W_i are the mass fraction and atomic mass of element i , with subscripts 1 and 2 representing the fuel and air streams, respectively.

$$\xi = \frac{(Y_H - Y_{H,2})/2W_H - (Y_O - Y_{O,2})/W_O}{(Y_{H,1} - Y_{H,2})/2W_H - (Y_{O,1} - Y_{O,2})/W_O} \quad (3)$$

After the grid independence verification, a total of 4.15 million grid cells are used to discretize the computational domain. Stretched grids are employed, with the grid size increasing along the radial direction. The grid near the shear layer is refined to resolve the interaction between the jet and pilot. A second-order scheme is used for spatial discretization, and a first-order Euler scheme is employed for time advancement. The development of turbulent structures is achieved through 5 flow-through times in the simulations, followed by an additional 10 flow-through times to obtain time-averaged and root mean square (rms) statistics. 512 CPU cores are used for the computation, with a total of 205 and 350 h spent on Flame D and Flame F, respectively.

5. Results and discussion

In this section, the instantaneous temperature distributions of Flame D and Flame F are first presented to qualitatively analyze the transient structure of the flames. After that, the simulation results of LES-DMC for Flame D and Flame F are quantitatively compared with experimental data. Since LES provides Favre-filtered quantities, experimentally measured data employing Favre averaging are used to ensure consistency. The data used for comparison are sampled from cross-sections at different axial positions to comprehensively evaluate the performance of LES-DMC throughout the entire computational domain. Finally, the prediction of local extinction is analyzed based on the distribution of scalar dissipation rate.

5.1. Instantaneous flame structure

The instantaneous temperature distributions of Flame D and Flame F are shown in Fig. 3. In order to facilitate the splicing of shapes, the cross-section of the computational domain is trimmed to be of equal diameter. It can be observed that the high-temperature pilot flame quickly ignited the surface layer of the jet. Meanwhile, the area near the axis maintained a low temperature until the fresh fuel/air mixture at the center of the jet begin to burn in the downstream region. Under the action of shear stress, turbulent structures form rapidly. The intense fluctuations cause the flame wrinkling. With a higher Reynolds number, the flame front in Flame F wrinkles earlier compared to Flame D. Moreover, it can be observed that the high-temperature region (around 2000 K) in Flame D locally becomes thinner, such as at

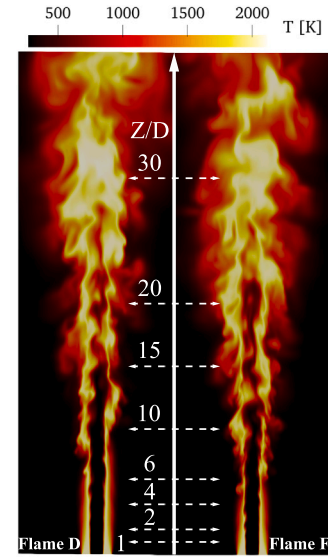


Fig. 3. The instantaneous temperature distributions of Flame D and Flame F.

$Z/D = 8$, but remains continuous. In contrast, the thin layer appears earlier in Flame F (at $Z/D = 4$), and even discontinuities occur at $Z/D = 8$ and 11. A sharp temperature drop and discontinuous flame structure typically signify local extinction, implying that the simulation captured this phenomenon. Furthermore, Flame F exhibits stronger local extinction compared to Flame D, which is consistent with the experimental observation.

5.2. Comparisons in physical space

In this subsection, comparisons between the simulation results and experimental measurements in physical space are performed for both Flame D and Flame F. The radial distributions of temperature, mixture fraction, and OH mass fraction at four different axial locations ($Z/D = 1, 4, 10, 20$) are presented in Fig. 4. Due to space constraints, the distributions of major species are provided in the supplementary material. Additionally, to avoid significant overlap with the mean values that would reduce readability, the rms distribution of the OH mass fraction is also separately provided in the supplementary material. The results of Flame D shown in Fig. 4(a) are discussed first. At the location near the nozzle where $Z/D = 1$, the simulated temperature peak and profile are very close to the experimental ones, indicating that the settings for the pilot temperature and the size of the inlets are reasonable. Beginning at $Z/D = 4$, it is observed that the temperature peak slightly decreases compared to that at $Z/D = 1$, while the temperature near the jet gradually increases along the downstream direction. This is due to shear-induced mixing between the pilot and the jet, which ignites the mixture within the jet. It is worth noting that the increase of the simulated temperature near the center-line shows a slight delay compared to the experimental results, indicating that the turbulent mixing in the simulation is slightly weaker than that in the experiment. This is also supported by the comparison of the rms values. At the upstream positions of $Z/D = 1$ and 4, the rms values of temperature are underestimated, suggesting insufficient local fluctuations. Nevertheless, at the downstream positions of $Z/D = 10$ and 20, the simulation accurately reproduces the fluctuation phenomena observed in the experiment. For the mixture fraction, the simulation predictions and experimental measurements show a high degree of consistency, with only a slight underestimation of the rms value at the upstream locations of $Z/D = 1$ and 4, which has the same cause as the underestimation of the temperature rms values. The good agreement of the mixture fraction distribution demonstrates that the

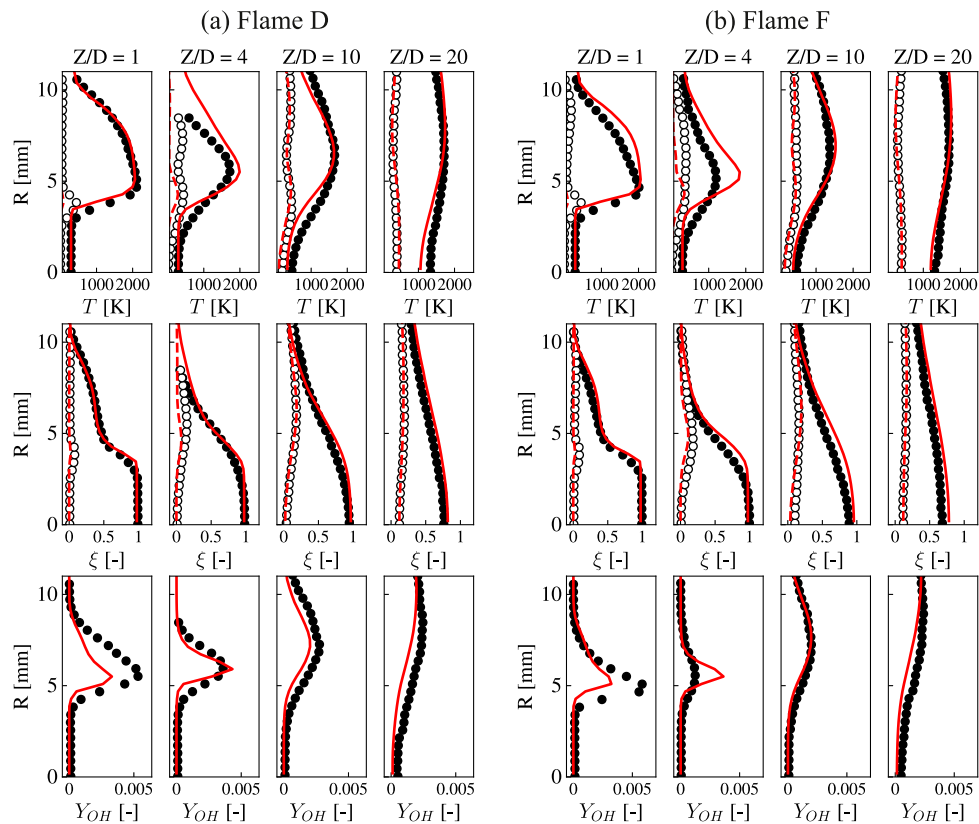


Fig. 4. Distributions of temperature, mixture fraction, and OH mass fraction in physical space for (a) Flame D, (b) Flame F (solid line: simulated time-averaged value, dashed line: simulated rms value, filled circle: experimental time-averaged value, circle: experimental rms value).

mixing process of fuel and oxidizer in the experiment is well simulated. The mass fraction of OH is underestimated at the location near the nozzle where $Z/D = 1$, while the prediction of OH is significantly improved downstream. The underestimation near the nozzle may be due to the following reasons: First, the region where OH is underestimated is downstream of the pilot inlet ($R = 3.75 \text{ mm} - 9 \text{ mm}$). Since the pilot consists of burned gases and the location of $Z/D = 1$ is very close to the inlet, the OH peak at this location is largely dependent on the pilot inlet composition. The pilot composition obtained from adiabatic calculation in this work may differ from that of the experiment, which is particularly evident for minor species like OH. Second, turbulence is not yet fully developed in the region near the nozzle, and molecular diffusion plays a significant role in transport. Therefore, differential diffusion effect has a significant impact in the near field, especially in hydrogen-enriched flames, as previously reported in studies [27–29]. This indicates that the unity Lewis number assumption adopted in the present work may introduce certain deviations. Overall, the LES-DMC has successfully reproduced the thermo-chemical state of Flame D, especially in the midstream and downstream regions, where it achieves good agreement with experimental measurements.

The results for Flame F are presented in Fig. 4(b). Overall, the discrepancies between the simulation results and experimental data are similar to those of Flame D, except at $Z/D = 4$. Looking at the time-averaged temperature, the experimental temperature peak of Flame F shows a significant decrease at $Z/D = 4$ compared to $Z/D = 1$, which is clearly different from Flame D. According to the experimental study [5], discontinuous flame structures were observed in the short exposure imaging of Flame F from 2D to 4D in height (see Fig. 1 of Ref. [5]), indicating the presence of local extinction. This was also confirmed by the subsequent probability density functions analysis (see Fig. 5 of Ref. [5]): At $Z/D = 4$, the temperature distribution of Flame D is concentrated around a high value of 2000 K, whereas the temperature distribution of Flame F encompasses a broader range from 500 K to

1900 K. Therefore, the significant local extinction in Flame F results in a lower time-averaged temperature at $Z/D = 4$, which the simulation in this work failed to capture adequately. Due to the same reason, the simulation overestimates the OH mass fraction at the height of $Z/D = 4$. Since the mixture fraction is independent of the reaction progress, it is minimally affected by local extinction. Overall, the predictions of LES-DMC produce deviations from experimental measurements in the upstream region where the local extinction occurs, but consistency with the experimental data is still maintained in the midstream and downstream regions.

5.3. Comparisons in mixture fraction space

Comparisons in the mixture fraction space can reveal the differences in the reaction process between simulation and experiment. Here, the data used for comparison were sampled from 100 snapshots within 0.002 s after the flame reached statistical steady state. The distributions of temperature, OH mass fraction in mixture fraction space at four different axial locations ($Z/D = 1, 4, 10, 20$) are shown in Fig. 5. For Flame D, the mean temperature distribution from the LES-DMC simulation almost coincides with the experimental data, with only an underestimation of fluctuations on the fuel-rich side, as presented in Fig. 5(a). This indicates that in regions with the same mixing conditions, the simulation accurately reproduces the combustion state observed in the experiment. The OH mass fraction is underestimated at the upstream locations $Z/D = 1$ and 4 in the simulation. Similar to the analysis of the results in physical space, the OH near the nozzle is influenced by the initial OH concentration of the pilot, and preferential diffusion also plays an important role. Nevertheless, the predicted OH distribution along the downstream direction gradually approaches the experimental one, almost coinciding at $Z/D = 10$ and 20. Overall, the predictions of LES-DMC for Flame D in the mixture fraction space agree well with experimental measurements.

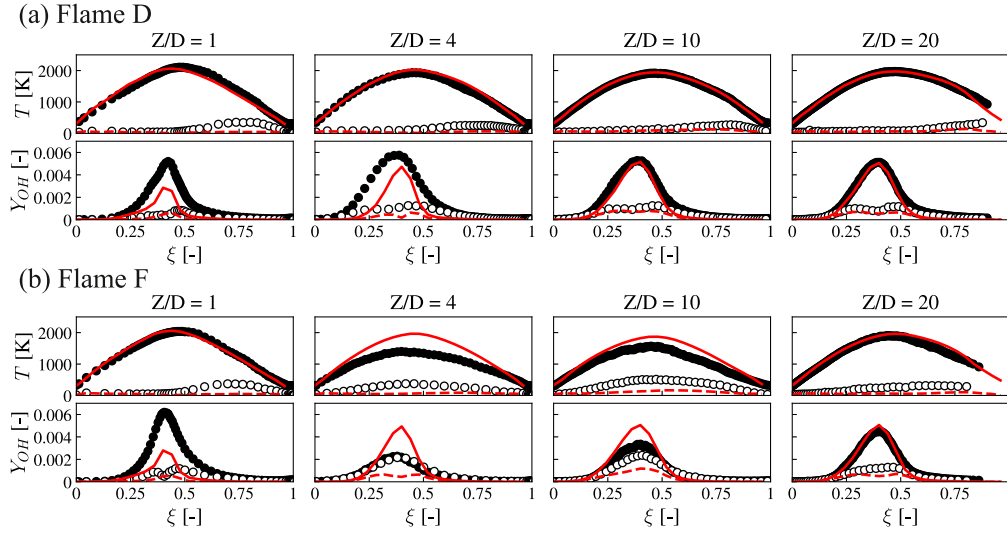


Fig. 5. Distributions of temperature and OH mass fraction in mixture fraction space for (a) Flame D, (b) Flame F (solid line: simulated conditional mean value, dashed line: simulated conditional rms value, filled circle: experimental conditional mean value, circle: experimental conditional rms value).

The results for Flame F are displayed in Fig. 5(b). The simulated and experimental temperature distributions agree well at $Z/D = 1$, but a significant deviation occurs at the subsequent location $Z/D = 4$. It is known that strong local extinction was observed at $Z/D = 4$ in the experiment. Local extinction leads to the interruption of reaction progress, resulting in greater fluctuations and lower mean values in temperature under the same mixture fraction conditions. However, the simulation underestimates the fluctuations and overestimates the mean values, suggesting an underestimation of the local extinction level. This effect continues until $Z/D = 10$, keeping the temperature peak elevated. However, this discrepancy gradually diminishes along the flow direction, and at $Z/D = 20$, the simulation and experiment are once again in agreement. A similar phenomenon is observed in the comparison of OH mass fraction. As an intermediate species, OH is concentrated in the active reaction zone. The inadequate prediction of local extinction results in an overestimation of the OH mass fractions at $Z/D = 4$ and $Z/D = 10$. However, with the combustion stabilizing downstream, the OH distribution is accurately reproduced at $Z/D = 20$. This trend is related to the variation of the extinction probability along the axial direction in Flame F. As shown in Fig. 4 of Ref. [5], the extinction probability reaches its peak around $Z/D = 4$ and remains at a high level until $Z/D = 10$, after which it rapidly decreases. The comparison for Flame F shows that the current LES-DMC simulation fails to accurately predict the phenomenon of local extinction. Nevertheless, the high consistency downstream indicates that LES-DMC is still accurate outside the local extinction region.

5.4. Flame extinction under intense mixing

In turbulent flames, local extinction can occur when excessive mixing rates cause diffusive heat losses to surpass local heat release. Fig. 6 displays instantaneous distributions of scalar dissipation rate to clarify mixing behavior in Flames D and F. Stoichiometric mixture fraction ξ_{st} iso-lines are also included to identify the reaction core zone, along with 25% peak OH iso-lines for observing local extinction [30]. It can be observed that in the upstream region with $Z/D < 4$, the peak scalar dissipation rate is located at the fuel-rich side, with a physical separation between ξ_{st} and high dissipation regions. Therefore, stable combustion is maintained as turbulent mixing contributes insignificantly to heat loss within the reaction core zone. However, in the downstream regions with $Z/D > 4$, partial overlap emerges between the ξ_{st} iso-lines and high dissipation zones. In regions where ξ_{st} and high dissipation zones

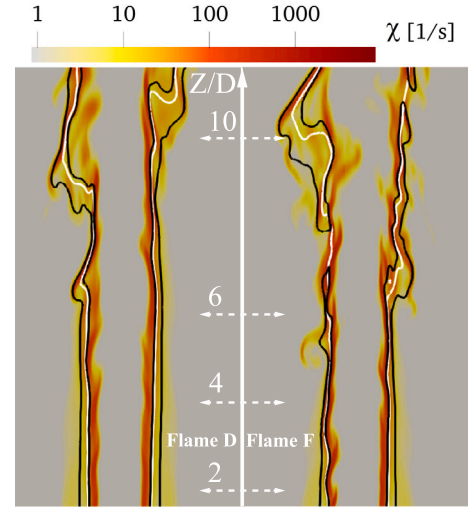


Fig. 6. The instantaneous scalar dissipation rate distributions of Flame D and Flame F (white solid line: stoichiometric mixture fraction iso-line, black solid line: 25% peak OH iso-line).

overlap, the OH iso-line becomes thinner and even exhibits localized voids, indicating the occurrence of local extinction. In the experimental study [5], Flame F exhibited the highest extinction probability at the 4D height. However, the simulation results show that the extinction intensity at $Z/D = 4$ is even lower than in downstream regions, suggesting an underestimation of extinction level. This discrepancy may be attributed to multiple factors. First, given the observed underestimation of upstream rms values in physical space comparisons, the turbulent inlet fluctuations in the simulation may be weaker than in experiments, which could consequently lead to reduced turbulence mixing level. Second, the algebraic closure used in the DMC model for fluctuation terms is derived by analogy with the sub-grid covariance model of two conserved scalars [22], which relies on the assumption of local equilibrium. It assumes a balance between production and dissipation rates at the subgrid scale, which may not be valid in the extinction condition. Additionally, the potential influence of differential diffusion on local extinction was not considered in the present work.

6. Conclusions

In this work, large eddy simulations of the KAUST piloted partially cracked ammonia jet flames were conducted using the direct moment closure model. At four different axial locations, quantitative comparisons were made between model predictions and experimental measurements in physical space and mixture fraction space for both Flame D and Flame F. Overall, the results of LES-DMC show good agreement with the experimental data, demonstrating the potential of DMC in predicting turbulent partially cracked ammonia combustion. However, in the prediction of Flame F, a significant deviation was observed at the height of $Z/D = 4$. This indicates that the LES-DMC performed in this work cannot fully reproduce the local extinction in Flame F. Extinction is a strong unsteady process, and the error introduced by closing second-order correlation terms with algebraic models may be amplified when faced with such phenomena. This emphasizes that accurately capturing local extinction in high Reynolds number turbulent flames remains a challenge in turbulent combustion modeling. How to improve the modeling to enhance the predictive capability for unsteady phenomena is a focus for future study.

Novelty and significance statement

The main novelty of this research is that (1) this work further accesses the performance of the direct moment closure (DMC) model, which directly close the chemical source term and does not need to introduce any assumptions of the combustion mode, for turbulent partially cracked ammonia flames using large eddy simulation (LES); (2) detailed comparisons and analyses are performed with the experimental data of target flames in both the physical and mixture fraction spaces. This study is significant because (1) it broadens the validation scope of the DMC model from previous hydrocarbon fuels to carbon-free fuels; (2) by focusing on assessing the performance in local extinction areas, it highlights the importance of considering unsteady effects in turbulent combustion modeling.

CRedit authorship contribution statement

Xin Zhou Tang: Writing – original draft, Methodology, Investigation. **Jiangkuan Xing:** Writing – review & editing, Formal analysis, Conceptualization. **Yicun Wang:** Investigation. **Qingqing Xue:** Data curation. **Kun Luo:** Writing – review & editing, Supervision, Project administration, Conceptualization. **Jianren Fan:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.proci.2025.105927>.

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